Dog Breed Recognition System

Group 27 (Null)

GITHUB: https://github.com/Nann1207/Null

YOUTUBE LINK: https://youtu.be/Ic4UQqTkkCY

Group Members and Contributions

100 %

Joshe D/O Chantiramugan

UOW ID: 8451242
Tutorial Group : T05
Contribution: Introduction and Results

100 %

Niruba Annriea Kichor Sagayaradje

UOW ID: 8560249
Tutorial Group : T05
Contribution: Python Scripts for all 3 models

100 %

Keeve Wong Ye Kai

UOW ID: 8865425
Tutorial Group : T05
Contribution: Conclusion & Troubleshooting

Teng Wei Chee

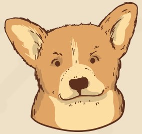
UOW ID:7771939
Tutorial Group : T04
Contribution: Methods

100 %

Wong Poh Yee

UOW ID: 8561588
Tutorial Group : T05
Contribution: Results

100 %



Vania Graciella Kwee

UOW ID: 8577201
Tutorial Group : T05
Contribution: Literature Review, Python Scripts & Readme

100 %

Introduction

Project Overview: This project focuses on developing a dog breed recognition system using deep learning models. The primary objective is to classify different dog images into their respective breeds. Distinguishing between different dog breeds can be challenging, especially when dogs share similar physical characteristics. With the advancement of deep learning, transfer learning using pre-trained models provides an effective approach to tackle this challenge.

Background: The traditional way of identifying dog breeds rely on humans expertise or manual classification which can take a lot of time and it's more prone to errors. The field of image classification has evolved greatly with deep learning. One model that automates feature extraction and improves classification accuracy is CNNs(convolutional neural networks). Pre-trained CNN models such as ResNet50, VGG16 and InceptionV3 are models that have provided excellent performance in various image classification task.

Aims: Our main goals are:

- To implement and evaluate a few existing deep learning models for the dog recognition system.
- Compare the different models such as ResNet50, VGG16 and InceptionV3 and evaluate the best performing model with the highest accuracy.
- Identify the most accurate model for the image classification system.

Significance: Recognizing dog breeds has various applications, including pet identification, veterinary diagnostics, and animal conservation. This project advances the field of computer vision and deep learning by evaluating different architectures to determine the most effective model for breed classification. Furthermore, it offers valuable insights into transfer learning techniques for image recognition.

Literature Review

Scope

- Focuses on automated dog breed classification from images
- Reviews existing CNN architecture and transfer learning in image classification tasks
- Exploration of feature-based and end-to-end learning approaches

Survey of Techniques, Approaches, and Methods:

- Techniques: Baseline CNN and Usage of pre-trained models → ResNet50, VGG16, InceptionV3
- Performance Metrics: Accuracy, precision, recall, f1-score, computational efficiency

Techniques	Advantages	Disadvantages
Baseline CNN	Fast training, low computation	Poor accuracy with complex patterns, not practical
ResNet50	Deep, skip connections prevent vanishing, strong with complex images	High computational cost, slow, need large dataset
VGG16	Simple architecture, good at transfer learning, commonly used	Large model, slow inference, high memory usage
InceptionV3	Efficient multi-scale feature learning, good speed, high accuracy	Complex architecture, harder to implement

Contextual Analysis

- Builds on previous research, comparing multiple CNN architectures on the same dataset
- Investigates strengths and weaknesses for each breed recognition techniques
- Aims to enhance model generalisation using data augmentation

Method

Approach Overview:
We use three pre-trained models—InceptionV3, ResNet50, and VGG16—to identify the best performer for dog breed classification. This strategy allows us to leverage unique model strengths and ensure the highest accuracy.

Design and Implementation:

- **Data Preprocessing -** We apply standardization and augmentation (rotation, zoom, flip) to combat overfitting and enhance model generalization.
- **Model Selection -** Our choice of models includes VGG16, InceptionV3, and ResNet50, each adapted with custom layers to effectively identify a diverse range of dog breeds.

Implementation Details:

- **Architecture -** Utilize pre-trained CNNs (InceptionV3, ResNet50, VGG16) with additional dense layers and dropout for dog breed classification, leveraging ImageNet weights.
- **Code Structure -** Implemented in Python using TensorFlow and Keras, ensuring robust training and evaluation.
- **Third-Party Code and Modifications -** Our models, adapted from Ayush Dabra's GitHub repository, include tailored enhancements for dog breed classification. Specifically, InceptionV3 and ResNet50 use frozen original layers with new top layers trained, while VGG16 features extensive fine-tuning of deeper layers and robust data augmentation to improve generalization.

Method

Innovation:

- **Efficient Feature Extraction -** Utilizing CNNs for their superior ability to autonomously extract essential features from image data, crucial for distinguishing dog breeds based on shapes and textures.
- **Learning Rate Strategy:**
    - **Initial Settings -** Starting with tailored learning rates (0.0001 for ResNet50 and InceptionV3; different for VGG16) to ensure detailed parameter optimization.
    - **Dynamic Adjustment -** Employing ReduceLROnPlateau to adjust the learning rate based on validation metrics, enhancing model stability and performance.

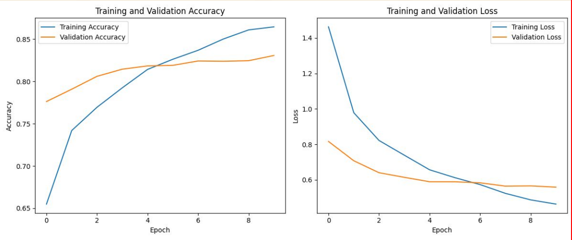
Key Applications

- **Transfer Learning -** Accelerates the training process using models pretrained on extensive image datasets, improving time-to-accuracy.
- **Fine-tuning -** Refines sensitivity to breed-specific features, increasing the accuracy in identifying similar breeds.
- **Data Augmentation -** Crucial for training robustness, expanding data variability to enhance performance on new, unseen images.

Citation: Adapted from Ayush Dabra's InceptionV3 for Stanford Dogs Dataset Classification. https://github.com/ayushdabra/stanford-dogs-dataset-classification/blob/master/inceptionV3-for-stanford-dogs-dataset.ipynb

Method

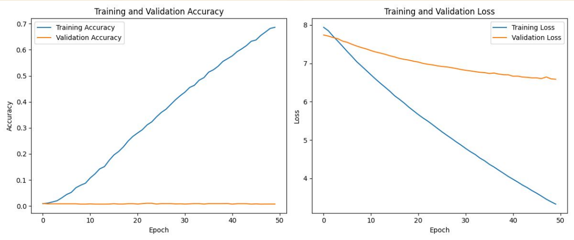
InceptionV3



Classification Metrics:
Accuracy: 0.8283
Precision: 0.8332
Recall: 0.8283
F1-score: 0.8273

Best performer with high stability in its loss and accuracy curves, suggesting excellent generalization abilities from its complex, layered architecture.

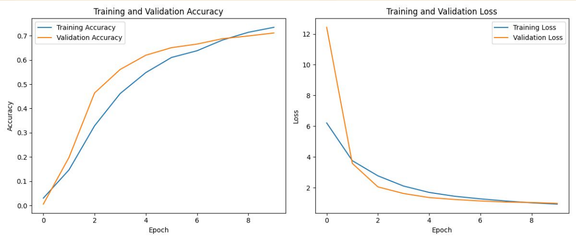
VGG16



Classification Metrics:
Accuracy: 0.3780
Precision: 0.4623
Recall: 0.3780
F1-score: 0.3695

Lower performance with a marked accuracy drop in validation, indicating overfitting due to a simpler, deeper network unable to handle complex patterns.

ResNet50



Classification Metrics:
Accuracy: 0.7211
Precision: 0.7316
Recall: 0.7211
F1-score: 0.7186

Moderate success, displaying smoother and more consistent training and validation curves thanks to its residual blocks that mitigate deep learning challenges.

Results - Experimental Setup & Evaluation Metrics

Datasets:
The dataset used is called **Stanford Dogs Dataset**. It is a dataset from Stanford University that includes **20,580 photos** of more than **120 different dog breeds**. **CNN** is used for the dog breed models and **Transfer Learning** is used in the below **3 models: InceptionV3, ResNet50 and VGG16**

The **preprocessing steps** include **image resizing** to ensure uniform dimensions, **normalisation** using scaling values to ensure all variables have the same scale, improving the performance and stability of the model, **data augmentation** to artificially expand the dataset and enhance model generalisation, and **a train-test split** to evaluate the model's performance on unseen data.

Parameters:

- Batch Size = 32
- Number of classes/labels = 120
- Initial Learning Rate = 0.0001
- Fine-tuning Learning Rate = 0.00001
- Optimizers: Adam
- Loss Function: Categorical Cross-Entropy
- Epochs: 20 for initial training, 10 for fine-turning (ResNet50 and InceptionV3), 50 for VGG16v

Evaluation Metrics:

- **Classification matrix** is used to access the **performance of the models.**
- **Training and loss accuracy** is used to indicate if the model is **learning from the training data efficiently.**
- **Confusion Matrix** is used to evaluate our models by helping us see **which breed are the most confused by the model.**
- **Classifying images,** the images generated will give us an insight as to **what images are being predicted wrongly and what the model is bias to.**

Results - Performance Metrics

Quantitative Results (Inception V3)

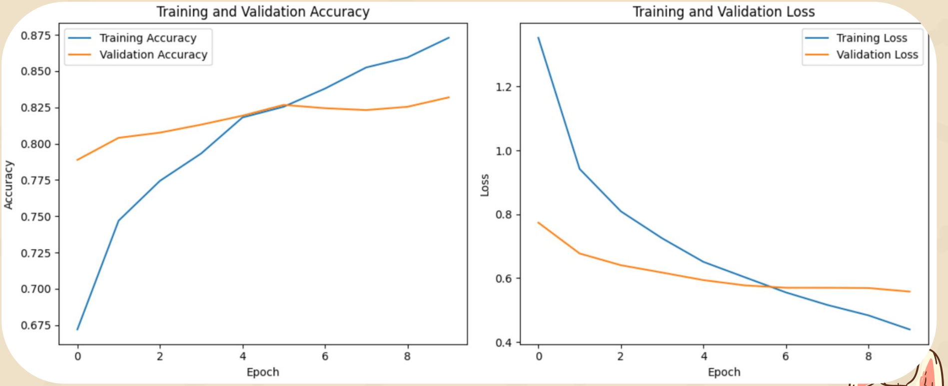
```
Classification Metrics:
Accuracy: 0.8283
Precision: 0.8332
Recall: 0.8283
F1-score: 0.8273
```

Accuracy of the model is is 82.83%, showing that the model has relatively high accuracy.
Precision of the model is 83.32%, showing that the model has high reliability.
Recall of the model is 82.83%, showing that the actual positives are well identified by the model.
F1-Score of the model is 82.73%, meaning that the model has balanced performance.

Training and Validation Accuracy

- **Training accuracy** is **steadily increasing** over the epochs, indicating that the model is learning from the training data.
- **Validation accuracy improves initially but dropped after a few epochs.**

This suggest a trend of overfitting as by the end of the training as the training accuracy is higher than the validation accuracy.



Training and Validation Loss

- **Training loss** is **steadily decreasing** over the epochs, indicating that the model is reducing error on the training data.
- **Validation loss decreased initially but slightly increase towards the end.**

This suggest a trend of overfitting as by the end of the training, the divergence between training and validation loss might indicative that the model is memorising the data rather than generalising well to the unseen data.

Results - Performance Metrics

Qualitative Examples (Inception V3)

Confusion Matrix(Summary)

```
Top Misclassified Dog Breeds:
Appenzeller misclassified as EntleBucher: 11 times
Siberian Husky misclassified as Eskimo Dog: 16 times
Miniature Poodle misclassified as Toy Poodle: 9 times
Eskimo Dog misclassified as Siberian Husky: 8 times

Most Confusing Pair of Dog Breed:
Siberian Husky misclassified as Eskimo Dog (16 times)
```

5 Random Classifier images

From evaluating the 5 random images some misclassification tendencies and bias was observed:

Confusion between similar breeds due to image clarity, angle, poses and other objects:

- Eg, Flat-coated Retriever was misclassified as Standard poodle, with the pose and coat of the dog not visible, a curly coat confused the model to predict it as a poodle. Hence clarity of image affected the classification
- Eg, Whippet was misclassified as Saluki, both the dog breed has a slender build and is only distinct from its facial features where a saluki has a more furry ear. Both breeds look similar but due to the image angle and crop the ears is not closely shown. Hence image angle and size affected the classification.
- Eg, Boston Bull and French bulldog were misclassified, they share similar features the distinct feature are their ears where boston bull ears are more angled and pointed. In the image there was a lack of clarity, which caused this misclassification.
- Eg, Eskimo Dog classified as Malamute, having a human in the image also affected the classification as the human in the image might distract the model from properly analyzing the dog's unique features
- Therefore, factors like angle, pose, clarity of images and having humans in the image could cause confusion while classifying the images causing the model to confuse them with similar breeds.

Results - Discussion & Improvement Ideas

Discussion:

Successes and Failures:

- **Successes:** The model performs well on most breeds, **achieving high accuracy and F1-scores**. The **use of transfer learning** (InceptionV3, VGG16, ResNet50) and **data augmentation** significantly improves the model's ability to **generalise to unseen data.**
- **Failures:** Some breeds are **frequently misclassified,** especially those with **similar visual characteristics** like breeds with similar coat colors or patterns. But this is highlighted by the most confusing pairs of breeds being identified.

Satisfaction with Results:

- The results are generally satisfactory, with the model **achieving high accuracy and good generalisation**. However, reducing misclassifications between visually similar breeds and the runtime can be done to optimise the model.

Improvement Ideas:

- **Data Augmentation** can be done to **further augment the dataset with more diverse transformations** such as different lighting conditions etc to improve the model's robustness.
- **Class Imbalance** can be addressed by **oversampling under-represented** breeds or using class-weighted loss functions.
- **Experimen**t with other models to **improve performance.**
- **Fine-tune** by having **more layers of the base model or use a larger learning rate** during fine-tuning to better adapt the model to the specific dataset.

Conclusion

Summary of work

- Purpose: Develop a dog breed recognition system using deep learning models
- Methodology: Implemented and compared ResNet50, VGG16, and InceptionV3 models
- Key Findings: InceptionV3 performed best with 82.83% accuracy

Key Takeaways

- Transfer learning with pre-trained models significantly improved classification accuracy
- Data augmentation and fine-tuning enhanced model generalization
- Similar-looking breeds posed the greatest challenge for accurate classification
- Mixed breeds will be a much harder problem to tackle

Broader applications

- Pet / breed identification accompanied with live camera, veterinary diagnostics

Thoughts

- This project advances computer vision applications in animal recognition, demonstrating the potential of deep learning in solving complex classification tasks.

Limitations & Future directions

Inaccurate classifications

- Some dog breeds look very similar, which will lead to misclassification e.g. Siberian Husky & Eskimo dog
- Having so many classifications (120) & insufficient samples (avg 172 samples / classification) can lead to inaccuracy
  - Having more samples per classification would boost the model's ability to analyze

Image quality & artifacts

- Unwanted artifacts - Having humans or the background in the image skewing the model's analysis
  - Remove humans / background or unwanted artifacts from images except the dogs
- Lack of image quality can cause the model to perform sub optimally
  - Source and process images in higher definition with different lighting / angles / posing to avoid overfitting

Runtime

- Long run times due to high amount of images to process (20,580)
  - A stronger processor would lower run times

Real world applicability & scalability

- The model may not perform as well with real-world images due to differences in lighting, background, or image quality
- Might not perform well on mixed-breed dogs in real world application, as it is trained only on purebred images